

Vision-Guided Multi-Objective Reinforcement Learning for UR5 Manipulation

Project Overview and Expected Results

Research proposal summary

Abstract

Real-world robot manipulation requires a robot to achieve its task while balancing conflicting objectives such as task success, energy consumption, movement smoothness, and safety. The literature provides each of the required capabilities — vision-guided control, multi-objective reinforcement learning (MORL), 6-DoF manipulation, preference conditioning, and safety as a constraint — but no existing system combines all of them on an articulated arm operating in dynamic scenes. This project proposes a preference-conditioned, vision-guided multi-objective policy for a UR5 robot, built on a 3D point-cloud representation, a multi-objective critic, a diffusion-based visuomotor policy, and a constrained-MDP formulation of safety. The expected outcomes, including a target of at least 70% static success on the real robot and a hypervolume at or above existing MORL baselines, are hypotheses to be confirmed by experiment rather than measured results.

1 Introduction and Motivation

The immediate stimulus for this project is that two otherwise mature research lines — multi-objective reinforcement learning and vision-guided manipulation — have developed largely independently. MORL methods provide principled tools for balancing several rewards, while visuomotor policy learning has delivered strong results on contact-rich arm control. What is missing is their integration into a single system with the following five properties:

1. **Vision:** RGB-D / point-cloud observations of the scene.
2. **MORL:** vector rewards and a Pareto-optimal solution set.
3. **Manipulation:** contact-rich, 6-DoF arm control.
4. **Preference conditioning:** runtime modification of behavior via a preference vector λ , without retraining.
5. **Safety as a constraint:** collision, velocity, and workspace limits expressed as a constrained MDP (CMDP), separate from the preference trade-off.

The objective is to obtain one trained policy that observes the scene, acts on the arm, and whose trade-off among success, energy, and smoothness can be changed at run time — all while respecting hard safety limits.

2 Literature Review

The review summarizes the relevant methods in five thematic threads and then identifies the research gap. Numerical values quoted below are those reported by the cited publications.

2.1 Multi-Objective Reinforcement Learning

The MORL literature supplies the algorithmic basis for policies that balance multiple objectives.

Envelope MOQ [1] trains a single network $Q(s, a, \omega)$ over the preference space. Its envelope Bellman operator and hindsight experience replay provide the foundation for single-policy preference conditioning.

PGMORL [2] maintains a population of PPO policies trained under different preference vectors, guided toward the Pareto front by a Gaussian-process prediction model. Solutions are evaluated by hypervolume and sparsity.

MO-MPO [3] replaces direct scalar weighting with per-objective KL budgets, yielding a scale-invariant treatment of multiple objectives.

DOL [4] demonstrates multi-objective learning with image observations in the ImageDeepSea environment; the visual setting is a toy grid rather than a camera-based manipulation problem.

MO-Gymnasium [5] provides the standard vector-reward interface for multi-objective environments, which the proposed benchmark adopts.

Limitation. These methods form the MORL foundation, but none simultaneously addresses visual 6-DoF manipulation in dynamic scenes with explicit safety constraints.

2.2 Vision-Guided Robot Manipulation

Vision-guided manipulation has advanced rapidly through imitation and visuomotor policy learning.

Diffusion Policy [6] represents the policy as a conditional diffusion process over action sequences. Its reported UR5 setup uses $T_o = 2$, $T_p = 16$, and $T_a = 8$, a 10 Hz policy with 125 Hz interpolation, a Cartesian velocity limit of < 0.43 m/s, and a workspace position at least 1 cm above the table.

3D Diffusion Policy (DP3) [7] learns manipulation from sparse point clouds of 512 or 1024 points, reporting a 24.2% relative gain over image-based baselines across 72 simulation tasks and 85% real-robot success with 40 demonstrations.

ACT / ALOHA [8], **Mobile ALOHA** [9], **UMI** [10], and **IRIS** [11] are further demonstration-based visuomotor manipulation systems.

Limitation. These approaches primarily optimize a single task objective and provide no runtime mechanism for changing the trade-off among multiple objectives.

2.3 Robot-Scale Multi-Objective RL

GCR-PPO [12] is the closest algorithmic analogue of the proposed multi-objective critic. It uses a multi-head critic $V_\phi : S \rightarrow \mathbb{R}^K$ with component-wise generalized advantage estimation, ratio-preserving advantage normalization, and asymmetric PCGrad. The method reports a $\sim 4.55\%$ average return gain over PPO, with gains up to 9.5% on multi-objective suites.

These values are treated in this project only as an *algorithmic reference* for the order of magnitude of improvement; they are not adopted as numerical targets for the UR5 system. GCR-PPO operates on proprioceptive observations and does not combine a visual encoder, runtime preference conditioning, or CMDP safety.

2.4 Safety-Constrained MORL

PRC-CMORL [13] formulates multi-objective robotic control as a constrained MDP. Its key distinction is that a *safety objective* is not the same as a *safety constraint*: if safety enters only as a reward term, the policy

may trade safety away for performance when λ changes. A CMDP instead requires the safety limits to be satisfied explicitly, for example

$$d(s, O) \geq D_{\text{safe}}, \quad \|\dot{x}\| < 0.43 \text{ m/s}. \quad (1)$$

In the proposed system the preference vector therefore controls only the task-objective trade-off among success, energy, and smoothness, while safety is handled separately as a constraint.

2.5 Closest Intersections: Vision and MORL

Demonstration-enhanced adaptable multi-objective navigation [14] is the closest combination of MORL, preference conditioning, dynamic environments, and sim-to-real transfer. It employs a preference interpolator $I(\lambda) = \lambda^p$ and adapts the preference at runtime without retraining. However, it uses lidar-based wheeled navigation rather than visual 6-DoF manipulation, and collision is treated as a sparse objective rather than a CMDP constraint.

POVNav [15] combines Pareto reasoning with visual input and explicit safety constraints, but its focus is classical navigation planning rather than a learned preference-conditioned manipulation policy.

2.6 Research Gap

The literature thus supplies the individual building blocks described above, while the combined system remains unaddressed:

Vision + MORL + 6-DoF Manipulation + Preference Conditioning + CMDP Safety

on a UR5 arm in static and dynamic environments. The following summary tables make the gap explicit. Capability comparison:

Method	Vision	MORL	Dynamic	Manip.	Preference	Safety
Envelope MOQ (2019)	×	✓	×	×	✓	×
PGMORL (2020)	×	✓	×	×	✓	×
MO-MPO (2022)	×	✓	×	partial	✓	×
Diffusion Policy (2023)	✓	×	✓	✓	×	partial
DP3 (2024)	✓	×	✓	✓	×	empirical
ACT / ALOHA (2023)	✓	×	✓	✓	×	×
GCR-PPO (2025)	×	✓	partial	✓	×	×
PRC-CMORL (2023)	×	✓	×	×	✓	✓
MORL navigation (2025)	lidar	✓	✓	×	✓	×
POVNav (2025)	✓	classical	✓	×	partial	✓
This work	✓	✓	✓	✓	✓	✓

What each field provides and what remains:

Research area	What it provides	Remaining gap
MORL	Pareto optimization	Limited visual manipulation
Vision manipulation	Visual 6-DoF control	Mostly single-objective
GCR-PPO	Multi-objective critic	No visual preference-conditioned UR5
PRC-CMORL	Safety constraints	No visual manipulation
MORL navigation (2025)	Preference adaptation in dynamic scenes	Navigation rather than manipulation
POVNav	Visual Pareto planning + safety	Classical navigation planning
This work	Combines all five	UR5 dynamic manipulation

The proposed study therefore investigates this combination on a UR5 in static and dynamic scenes.

2.7 Positioning Relative to Vision-Guided Baselines

The vision-guided methods reviewed above are the *visual-manipulation foundation* (the adopted backbone) of the proposed system, while the MORL methods form its *algorithmic foundation*. The proposed approach does not reproduce their single-objective benchmarks; it evaluates on different criteria. This yields two distinct classes of claims.

Where the proposed approach can improve on these works:

- **Dynamic scenes.** The vision-guided works largely target static-scene manipulation. The proposed benchmark adds a moving obstacle, a moving target, and sudden appearance, which those methods do not handle as a learned behavior.
- **One model, many behaviors.** They return a single fixed trade-off. The proposed policy produces a Pareto set of behaviors and lets the user select among them at run time.
- **Runtime preference, no retraining.** None of the vision-guided methods offers preference conditioning; the proposed policy changes its success–energy–smoothness trade-off without retraining.
- **Explicit safety.** Their safety handling is empirical or partial. The CMDP formulation yields explicit guarantees, evaluated as measurable violations (static $\approx 0\%$).
- **Controlled attribution.** The proposed evaluation isolates the effect of the multi-objective critic, the preference input, and the safety constraint (C3), which the vision-guided works do not provide.

Where the proposed approach does not claim to outperform them:

- **Raw static success.** DP3 is adopted as the backbone, so beating its 85% real success is not claimed; the static target of $\geq 70\%$ is deliberately lower because the system also optimizes energy, smoothness, and safety.
- **Representation gain.** The +24.2% point-cloud advantage is inherited from the backbone, not improved.
- **Cross-benchmark comparability.** Different tasks, sensors, and data scale prevent a direct numerical comparison with their published numbers.

In short, the defensible claim is: *better under multi-objective, preference, and safety criteria on a dynamic benchmark — not higher single-objective success than the vision baselines.*

2.8 Overall Contribution: From MORL to This Work

The overall contribution of the paper is best explained as a chain of three stages: the field of multi-objective reinforcement learning, its robot-scale realization, and the proposed unification. Each stage contributes a capability and leaves a limitation that the next stage addresses.

Stage 1 — Contribution of MORL. The MORL field supplies the principled formulation of multi-objective control: vector rewards, Pareto-optimal solution sets, and preference-conditioned policies (Envelope MOQ, PGMORL, MO-MPO), together with a standard benchmark interface (MO-Gymnasium). *Limitation carried forward:* these results are demonstrated mostly in abstract or toy settings, without visual 6-DoF manipulation, dynamic scenes, or explicit constraints.

Stage 2 — Contribution of robot-scale MORL. This stage scales MORL to realistic robot-learning problems. GCR-PPO provides a multi-head critic (per-objective value heads, component-wise GAE, ratio-preserving advantage normalization, asymmetric PCGrad) with $\sim 4.55\%$ average and up to 9.5% return gains, and the IROS 2025 navigation work demonstrates runtime preference adaptation. *Limitation carried forward:* proprioceptive or lidar inputs only; no vision-guided manipulation; collision treated as an objective rather than a CMDP constraint.

Stage 3 — Contribution of this work. The paper unifies the two stages into a single system:

- **C1:** a preference-conditioned, vision-guided MORL policy — DP3 point-cloud backbone, multi-objective critic, and diffusion policy — with runtime λ and no retraining;
- **C2:** a multi-objective UR5 benchmark (static scene plus three dynamic scenes) wrapped in MO-Gymnasium;
- **C3:** a controlled evaluation that separates the effects of multi-objective learning, preference conditioning, the visual representation, and safety, together with an explicit sim-to-real study.

Limitation resolved: the UR5 static-and-dynamic manipulation setting that the earlier stages leave open.

The contribution chain is summarized below.

Stage	Contribution	Limitation carried forward
MORL field	Multi-objective formulation, Pareto sets, preference tools	Not demonstrated on visual robots
Robot-scale MORL	Critic/GAE/PCGrad machinery; runtime preference adaptation	No vision-guided manipulation; safety as objective
This work	Unify MORL + robot scale + vision + CMDP on a UR5	UR5 static–dynamic manipulation (the gap)

3 Proposed Method

1. **Visual representation.** A DP3-style point-cloud encoder reduces the RGB-D observation to 512 points. The policy is warm-started from 10–20 demonstrations per task.
2. **Multi-objective critic.** One value head per objective, trained with per-objective generalized advantage estimation, ratio-preserving advantage normalization, and asymmetric PCGrad to mitigate conflicting gradients.
3. **Preference-conditioned policy.** A diffusion-based visuomotor policy conditioned on the runtime preference vector $\lambda = (\lambda_{\text{success}}, \lambda_{\text{energy}}, \lambda_{\text{smoothness}})$, so the trade-off can be changed without retraining.

4. **CMDP safety.** Constraints $d(s, O) \geq D_{\text{safe}}$ and $\|\dot{x}\| < 0.43$ m/s are enforced separately from λ .
5. **Benchmark.** A new UR5 multi-objective benchmark following the MO-Gymnasium interface, with one static scene and three dynamic scenes (moving obstacle, moving target, sudden appearance), so that all baselines share the same interface and conditions.
6. **Evaluation.** Simulation and real-robot training are run in parallel so that the sim-to-real gap is measured explicitly for every reported metric.

4 Expected Results vs. Reported Results

The following table places the measured results reported in the literature beside the expected outcomes of this study.

Quantity	Reported result (measured, source)	Proposed target (expected)
Point cloud	512 or 1024 points (DP3 [7])	512 points
Demonstrations	10–20 per task (DP3 [7])	10–20 (warm-start)
Real-robot success	85% with 40 demos (DP3 [7])	$\geq 70\%$ (static target)
Point cloud vs. images	+24.2% relative (DP3 [7])	Representation choice, not a gain target
Control loop	125 Hz (Diffusion Policy [6])	125 Hz
Cartesian velocity	< 0.43 m/s (Diffusion Policy [6])	< 0.43 m/s (constraint)
Multi-objective gain	$\sim 4.55\%$ avg / $\leq 9.5\%$ max (GCR-PPO [12])	Reference only; we report our own values
Pareto quality	PGMORL / Envelope MOQ hypervolume [2, 1]	Hypervolume reaches or exceeds those baselines
Safety	CMDP constraint (PRC-CMORL [13])	Static violations $\approx 0\%$; dynamic lower than baseline
Preference change	Retraining or separate policies (baselines)	3–5 settings, same policy, no retraining
Simulation success	—	$\sim 92\%$ (simulation reference)

Note. Values in the “Proposed target” column are expected outcomes of this study and must be confirmed by experiment; they are not measured results. After the experimental phase, each expected value will be replaced by its measured counterpart.

5 Research Questions and Expected Outcomes

Question	Comparison	Expected outcome
Q1: Can the visual representation support learning?	Proposed system vs. single-objective DP3 / PPO	$\sim 92\%$ in simulation; $\geq 70\%$ real static success
Q2: Does the multi-objective critic help?	Multi-objective critic vs. scalarized (single-number) PPO	Higher hypervolume / better Pareto front; not a fixed percentage gain
Q3: Does runtime preference conditioning work?	Continuous λ vs. fixed λ	Same policy, measurably different behavior, no retraining
Q4: Does CMDP safety reduce violations?	CMDP vs. collision-as-reward	Static violations $\approx 0\%$; dynamic violations strictly below the reward baseline

6 Experimental Plan (UR5)

1. Control cadence of 125 Hz with the policy updated at 10 Hz and interpolated; Cartesian velocity capped below 0.43 m/s.
2. Three to five preference settings exercised on the real arm, without retraining between settings.
3. Real static success target of at least 70%.
4. Success, energy, smoothness, and safety violations reported separately for the real robot.
5. Sim-to-real differences reported per metric (success, energy, smoothness, safety).

7 Evaluation Criteria

- **Visual learning.** $\sim 92\%$ simulation and $\geq 70\%$ real static success.
- **MORL benefit.** Hypervolume above the scalarized baseline.
- **Runtime preference.** 3–5 λ settings produce measurably different behavior from a single trained policy.
- **Pareto quality.** Hypervolume at or above PGMORL and Envelope MOQ on the same benchmark.
- **Safety.** Static violations $\approx 0\%$; dynamic violations strictly below the collision-as-objective baseline.

8 References

References

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